Pandas (pd): Used for data manipulation and loading datasets (DataFrames).

NumPy (np): Used for numerical operations and handling multi-dimensional arrays.

Matplotlib (plt) & Seaborn (sns): Used for creating static and statistical visualizations like heatmaps and plots.

Scikit-Learn (sklearn): The primary library for machine learning (preprocessing, model building, and evaluation).

NLTK & re: Used for Natural Language Processing (cleaning text, removing stopwords, and lemmatization).

TensorFlow / Keras: Building Deep Learning models (ANNs, CNNs).

TfidfVectorizer: Turning words into numbers for the AI to understand.

AIPD Practical Exam Record

Experiment 1: Exploratory Data Analysis (Diabetes Prediction)

- **Model Used: None (Data Exploration & Visualization)**
- **Explanation:** This experiment focuses on **Exploratory Data Analysis (EDA)**. Instead of predicting a value, it uses statistical methods to understand the dataset's structure, identify correlations between health metrics (like Glucose and Insulin), and clean the data for future modeling.
- **Libraries Used:** pandas, numpy, seaborn, matplotlib.
- **Unique Terms for Viva:**
    - **EDA (Exploratory Data Analysis):** The process of summarizing the main characteristics of a dataset, often with visual methods.
    - **PairGrid:** A seaborn feature used to plot pairwise relationships across an entire dataset to see how variables interact.
    - **isnull().sum():** A command used to identify and count missing or "NaN" values in each column.

Experiment 2: Spotify Track Popularity Prediction

- **Model Used: ANN (Artificial Neural Network) / Sequential MLP**

- **Explanation:** This experiment uses a **Deep Learning** approach to predict the popularity of Spotify tracks. It builds a **Sequential model** with multiple **Dense (fully connected) layers** using the **ReLU** activation function to perform regression on numerical and encoded categorical features like artist popularity and followers.
- **Libraries Used:** pandas, numpy, tensorflow.keras, sklearn (LabelEncoder, StandardScaler).
- **Unique Terms for Viva:**
    - **Label Encoding:** Converting categorical text data (like artist names) into numerical format so the neural network can process it.
    - **StandardScaler:** Normalizing numerical data so that all features have a mean of 0 and a variance of 1, which helps the neural network converge faster during training.
    - **MSE (Mean Squared Error):** The loss function used to measure the average squared difference between the predicted and actual popularity scores.

Experiment 3: IMDB Sentiment Analysis

- **Model Used: LogisticRegression**
- **Explanation: Logistic Regression** is a classification model used here to predict a binary outcome (Positive or Negative sentiment). It works by calculating the probability that a specific movie review belongs to a certain class based on the words it contains.
- **Libraries Used:** re, nltk, sklearn (LogisticRegression, TfidfVectorizer), seaborn.
- **Unique Terms for Viva:**
    - **Stopwords:** Common words (like "the", "a") that are removed because they don't carry emotional weight for sentiment detection.
    - **Lemmatization:** Reducing words to their root form (e.g., "better" to "good") to ensure the model treats related words as the same feature.
    - **TF-IDF Vectorizer:** A technique that converts text into numbers by highlighting unique, important words while devaluing common ones.

Experiment 5: AI Product (Amazon & Fake News Detection)

- **Model Used: MultinomialNB (Naive Bayes)**

- **Explanation: Multinomial Naive Bayes** is a probabilistic classifier. It is "naive" because it assumes that the presence of one word in a document is unrelated to the presence of any other word. It is exceptionally fast and accurate for text-based classification like news or reviews.
- **Libraries Used:** sklearn (MultinomialNB, TfidfVectorizer, LogisticRegression).
- **Unique Terms for Viva:**
    - **Noisy Text:** Raw data containing HTML tags, typos, or slang that can lower model accuracy if not cleaned.
    - **Ethical AI:** The responsibility to ensure news detection is accurate to prevent the accidental suppression of real information (Misinformation Suppression).

Experiment 6: Spam Detection & Movie Recommendation

- **Model Used: MultinomialNB** (for Spam) and **Cosine Similarity** (for Recommendation).
- **Explanation:**
    - **MultinomialNB:** Used to classify emails as "Spam" or "Ham" based on word frequencies.
    - **Cosine Similarity:** A mathematical metric used to measure how similar two movies are by calculating the "angle" between their feature vectors.
- **Libraries Used:** pandas, sklearn (MultinomialNB, TfidfVectorizer, cosine_similarity).
- **Unique Terms for Viva:**
    - **Encoding (latin-1):** A character encoding standard used to read files that might contain special symbols not found in standard UTF-8.
    - **Content-Based Filtering:** A recommendation strategy that suggests items similar to those a user liked in the past based on item features (tags, genre, etc.).

Experiment 8: Adversarial AI (Image Security)

- **Model Used: CNN (Convolutional Neural Network)**
- **Explanation:** A **CNN** is a deep learning model specifically designed to "see" and process image pixels. In this practical, we test the model's robustness against intentional digital attacks.
- **Libraries Used:** tensorflow, keras, numpy, matplotlib.

- **Unique Terms for Viva:**
    - **FGSM (Fast Gradient Sign Method):** An attack technique that adds a tiny, invisible layer of "noise" to an image to force the AI to make a wrong prediction.
    - **Adversarial Example:** An input to a machine learning model that has been intentionally designed to cause the model to make a mistake.

Experiment 9: Reddit Sentiment Analysis (WSB)

- **Model Used: LogisticRegression**
- **Explanation:** Used for **Multi-class Classification**. Unlike Exp 3, this categorizes text into three segments (Positive, Negative, or Neutral) to gauge the "mood" of a specific community like WallStreetBets.
- **Libraries Used:** pandas, sklearn (LogisticRegression, accuracy_score).
- **Unique Terms for Viva:**
    - **F1-Score:** A better metric than accuracy for imbalanced data; it combines Precision and Recall into one number.
    - **Classification Report:** A table showing the main classification metrics (Precision, Recall, F1) for each individual category.